

Ryley Nathaniel

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Objective — Seeking roles in healthcare or finance that combine statistical analysis with strategic communication.

Education

DOCTOR OF PHILOSOPHY (PhD) degree in PSYCHOLOGY **Expected August 2030**

York University, Toronto, ON

Supervisors: Dr. Liya Ma, Dr. Lara Pierce

Specialization: Brain, Behaviour, and Cognitive Sciences

- Research-based PhD thesis on error-related negativity (ERN) in the marmoset brain
- Connected Minds and VISTA trainee

MASTER OF ARTS (MA) degree in PSYCHOLOGY

Expected August 2026

York University, Toronto, ON

Supervisors: Dr. Liya Ma, Dr. Lara Pierce

Specialization: Brain, Behaviour, and Cognitive Sciences

- Research-based Master's thesis on decision-making neuronal behaviour in rats
- Connected Minds and VISTA trainee

DIPLOMAS

– Quantitative Methods Diploma

Requirements: completion of 18 credits with a minimum A- average, forum presentation, weekly forum attendance, statistics-focused practicum and research paper

– Graduate Diploma in Neuroscience

Requirements: completion of 6 credits, conference presentation, bimonthly talk attendance, pre-seminar question submission and neuroscience-focused research paper

– Collaborative Specialization in Vision Research

Requirements: completion of 6 credits and bimonthly talk attendance with pre- and post-seminar reflection submission and submission of 2 questions per speaker

HONOURS BACHELOR OF SCIENCE (BSc) degree in PSYCHOLOGY

December 2023

York University, Toronto, ON

Supervisor: Dr. Bradley Buchsbaum

Specialization: Cognitive Neuropsychology

- Key courses: Human Neuropsychology, Advanced Experimental Design and Psychobiology
- Led an Honours Thesis in creating and conducting an online psychology experiment using PsychoPy, analyzing data using Jamovi and writing a final APA-report

CERTIFICATES

- Certificate in Law and Social Thought
- Certificate in Sexuality Studies
- Certificate in Spanish Proficiency
- Certificate in Bilingual Excellence (French and English)

**completion of 18-credit requirement per certificate*

Awards/Scholarships

Connected Minds PhD Scholarship, York University – \$40,000; granted for 4 years of PhD research	2026-2030
CVR-VISTA Doctoral Entrance Award, York University (awarded, declined*) – \$10,000; offered for 1st year of PhD research – *Declined due to acceptance of Connected Minds PhD Scholarship	2026
CAN Professional Development Award – \$1,000 travel award	2026
Summer Graduate Fellowship, York University – \$3,602; 2 consecutive years	2025, 2026
Winter Graduate Fellowship, York University – \$3,602; 2 consecutive years	2025, 2026
Fall Graduate Fellowship, York University – \$3,602; 2 consecutive years	2024, 2025
Connected Minds Master’s Scholarship, York University – \$20,000; granted for 2 years of Master’s research	2024-2026
NSERC Undergraduate Student Research Award, York University – \$17,200 for decision making neuro research; selected 2 consecutive years	2023, 2024
Canada SROP Student Research Award, University of Toronto – 1 of 20 students selected for a paid research position (\$9,000)	2023
Visionary Talk Science Inspire Award, Visions of Science – 1 of 15 students selected to present on stage; winner of most inspiring presentation (\$200)	2022

Skills/Training

Languages English, French, Spanish, ASL, Portuguese	Soft Skills teamwork, problem-solving, leadership, creativity multi-lingual translation and interpreting, time-management, adaptability
Programming Software Python, R, PsychoPy, MATLAB Google & Microsoft Suites (e.g., xlsx, ppt, doc)	
Computational Neuroscience course completion, Neuromatch Academy	2025
MRI Level 1 Training, York University	2024
MoCA Test Administration Certification, MoCA Cognition	2023
Supported Conversation for Adults with Aphasia, Aphasia Institute	2023
Collaborative Institutional Training Initiative, Baycrest Centre	2022
TCPS 2: Core Completion, Panel on Research Ethics	2022

Research Experience

Research Assistant

York University, Toronto, ON

January 2023 – Present

Supervisor: Dr. Liya Ma

- Developing Python and MATLAB scripts for analyzing behavioural and neurological displays of cognitive flexibility in rats (using libraries such as Pandas, NumPy, SciPy, Scikit-learn, Statsmodels and MNE-Python), then visualizing data (using Matplotlib and Seaborn)
- Cleaning and preprocessing large datasets in Python, performing exploratory data analyses (EDAs) to identify trends and correlations in datasets
- Predicting rat decision-making response accuracy and reaction time using logistic regressions
- Plotting graphical representations of binary (accuracy) and non-binary (reaction time) data using Python (cumulative sum charts, histograms, and bar graphs)
- Automating reporting processes to improve efficiency and reduce manual data handling
- Provided one-on-one support to new coworkers (undergraduate volunteers)
- Authoring an APA-style manuscript, with both neural and behavioural data, for publishing
- Presenting in a soapbox style about cognitive flexibility for Visions of Science

Clinical Research Assistant

Baycrest Centre, Toronto, ON

August 2023 – April 2024

Supervisors: Dr. Angela Troyer, Dr. Komal Shaikh

- Translated two externally validated subjective memory assessment tools as part of a SPARK-CABHI collaboration to improve healthcare accessibility and broaden reach
- Validated the newly translated questionnaires in French, Spanish, Mandarin, and Cantonese
- Wrote APA research report of background, methods, and findings (potential for publication)
- Created survey using Microsoft Forms and advertised using physical posters in Baycrest

Research Assistant

University of Toronto Mississauga, Mississauga, ON

June 2023 – August 2023

Supervisor: Dr. Erika Carlson

- Employed R-programming to perform multilevel modeling on personality datasets
- Analyzed data collected by (a) self-reports, (b) informant surveys, and (c) EAR devices
- Wrote an APA research report

Research Assistant

University of Toronto Mississauga, Mississauga, ON

May 2023 – June 2025

Collaborator: Dr. Norman Farb

- Pioneered a study on the variance of interoception on the basis of location familiarity
- Outcome of networking through a Student Research Opportunity Program
- Underwent process of ethics and approval for research with human participants
- Coded a long-form baseline survey using Qualtrics for before and after intervention
- Coded a short survey using Qualtrics which participants were notified to complete throughout experimental days
- Collected data and handed it off to be analyzed as an Honours Thesis to undergraduate students at UofT

Research Assistant

University of Toronto Mississauga, Mississauga, ON

May 2023 – September 2023

Supervisor: Dr. Loren Martin

- Performed immunohistochemistry (IHC) tasks to analyze pain behaviour in mice
- Worked directly with tissue (staining, mounting, scoring)
- Learned how to use the cryostat machine for tissue slicing

- Administered morphine to mice and observed behavioural changes and drug dependencies

Honours Thesis Research Lead

*Rotman Research Institute, Toronto, ON
York University, Toronto, ON*

September 2022 – June 2023

*Supervisor: Dr. Bradley Buchsbaum
Supervisor: Dr. Buddhika Bellana*

- Developed an experimental memory task using PsychoPy and Python coding
- Analyzed data using Python and Jamovi to perform a three-way ANOVA
- Wrote an APA research report as part of the Honours Thesis course

Clinical Research Assistant

Baycrest Centre, Toronto, ON

June 2022 – June 2023

Supervisor: Dr. Komal Shaikh

- Co-authored an interactive, educational booklet using the platform Canva
- Conducting surveys to ask opinions of nurses, psychiatrists and therapists
- Planning an inpatient psychotherapy curriculum

Research Lead

York University, Toronto, ON

January 2022 – April 2022

Supervisor: Dr. Joey Cheng

- Collaborated with four other students to perform a laboratory experiment, delegating tasks amongst ourselves for an organized working model
- Promoted a survey to collect over 300 participants from all different backgrounds
- Presented findings at the 2022 Visionary Talk Science conference, earning the Visionary Talk Science Inspire Award for having the most "inspiring" presentation

Teaching Experience

Teaching Assistant (Psychology)

York University, Toronto, ON

September 2024 – April 2026

Course: PSYC 4175 – Advanced Community-Based Applied Research (prof: Dr. Michaela Hynie)

- Guided students through individual, community-based specialized honours thesis projects
- Lead three (3) tutorials, including one about the use of R for data analyses
- Attended every class to provide in-class office hours
- Marked students' methodological presentations and final APA-style theses

Course: PSYC 2130 – Personality (prof: Dr. Krista Phillips)

- Marked assignments and evaluations (2 midterms, 1 final exam)
- Provided one-on-one support to students who request office hours and exam review
- Invigilated evaluations/exams (midterm 1, midterm 2, and final exam)
- Lead two tutorials alongside a fellow teaching assistant, guided by our own Powerpoints

Course: PSYC 4180 – Critical Thinking in Psychology (prof: Dr. Edward Haltrecht)

- Marked assignments including presentations, proposals, SPSS write-ups and essays
- Provided one-on-one support to students who request office hour meetings
- Promptly responded to student emails asking about assignments or requesting extensions
- Communicated with the professor when students require extensions or additional support

Tutor**April 2022 – Present***Preply (online)*

- Teaching French, mathematics, science, and English to children aged 4-16
- Assisting children aged 4-16 with homework and assignments
- Assessing individual capacities of each child to ensure appropriate teaching methods
- Creating individual curriculums for each student depending on baseline skill-level and goals

Professional Experience

Insurance Bilingual Service Representative**April 2022 – April 2025***Empire Life, Kingston, ON*

- Audited sensitive health- and financial-related information containing hundreds of data points requiring transcription, validation and analysis
- Analyzed reports to decipher whether circumstances met criteria for certain coverages, then effectively communicated insights with colleagues and later, clients
- Facilitated French and English call-outs for life insurance clients
- Efficiently performed address, banking, beneficiary, power of attorney and child-rider changes in French and English
- Employed platforms such as OnBase, Ingenium, eCapsil and cCapsil

Medical Receptionist**April 2021 – April 2022***Westwood Health, Toronto, ON*

- Submitted extended health benefits to insurance providers through online portals (e.g., Greenshield & Sunlife), communicated with providers if issues emerged
- Opened and closed the location, in-charge of the cash (\$500), till, and patient scheduling
- Employed Jane App for medical records and Squarespace for website construction

Customer Service Representative and Opening/Closing Manager**June 2018 – June 2021***Dominos Pizza, Toronto, ON*

- Completed a General Manager in Training (GMIT) Level 1 course in June 2019
- Managed 15-20 people in a given shift (service representatives and delivery drivers)
- Opened and closed the location, in-charge of cash (\$1,000), till (\$100), and ordering
- Consulted with staff quarterly to go over monthly metrics demonstrating impact

Volunteer Experience

Executive Member at P.U.M.P.**September 2025 – Present***York University, Toronto, ON*

- Creating workshops to be used by the Psychology Undergraduate Mentorship Program (P.U.M.P.) program
- Social committee and outreach executive team member on behalf of PUMP at York University

Mentor at P.U.M.P.**September 2024 – Present***York University, Toronto, ON*

- Providing mentorship to undergraduate psychology students of York University

- Helping undergraduate students with CV building, email drafting, and job searching

Centre for Vision Research Retreat Planning Trainee

July 2025 – September 2025

York University, Toronto, ON

- Drafting and sending emails to 80 trainees, interactively sharing information about the retreat (outreach)
- Meeting with the rest of the planning team (8 members) and discussing next logical steps
- Collecting results from online Google Forms surveys about participants' interests and needs
- Effectively presenting participants' interests and needs from survey results to other planning team members in a digestible format (e.g., use of figures to demonstrate what most looking forward to)
- Transferring participant background data from Google Forms to Google Sheets for statistical breakdown
- Orchestrating conference Lightning Talk presentations for 11 speakers (email correspondence with instructions and reminders and timely PowerPoint slide compilation)
- Presenting and introducing each speaker for their Lightning Talks during the conference

Student Mentor

May 2023 – August 2023

York University, Toronto, ON

- Educated fellow undergraduate health students on the benefits of an experiential course

Communications Assistant

January 2023 – April 2023

Humber River Hospital, Toronto, ON

- Performed interviews to collect data to create a new hospital staff literature tool
- Co-authored an interactive training tool for new hospital staff

Conversation Workshop Specialist

September 2022 – September 2023

The Aphasia Institute, Toronto, ON

- Facilitated and advocated for accessible conversation for adults with aphasia
- Earned a certificate in Supporting Conversation for Adults with Aphasia

Theses

Mechanisms of performance monitoring: A marmoset model?	PhD thesis; Planned 2026
Medial frontal neuronal activities during repeated rule switches in rats	Master's thesis; 2024–Present
Measurements of self- and informant-knowledge	Research study; 2023–2024
Flexible decision-making in rats	Research study; 2023–2024
Does music jog associative memory?	Honours thesis; 2022–2023

Publications

- Rule-based flexible decision-making in rats [in-progress]** **Present**
Authors: Ryley Nathaniel, Yiping Zhang, Liya Ma — in-progress
- How to: Keep our minds at bay** **2024**
Authors: Ryley Nathaniel, Komal Shaikh — published privately at the Baycrest Centre Hospital
- New Staff Orientation Handbook** **2023**
Authors: Ryley Nathaniel, N. Morcos, I. Grisoni — published privately by Humber River Hospital

Poster Presentations

- Task rule representation in rat frontal cortical areas**
Authors: Ryley Nathaniel, Yiping Zhang, AbdulRafeh Usmani, Liya Ma
Canadian Association for Neuroscience (CAN) Conference, Montreal, QC 2026
CSBBCS Conference and CIAN-satellite, Toronto, ON 2026
- Medial frontal neuronal activities during repeated rule switches in rats**
Authors: Ryley Nathaniel, Yiping Zhang, Liya Ma
1. Society for Neuroscience (SfN) Conference, San Diego, USA 2025
2. York University CVR-CIAN Conference, Toronto, ON (abstract at <https://doi.org/10.25071/10315/42927>) 2025
3. Canadian Association for Neuroscience (CAN) Conference, Toronto, ON 2025
4. Natural Sciences and Engineering Research Council of Canada (NSERC), Toronto, ON 2024
- Neural mechanisms of performance monitoring: A marmoset model**
Authors: Ryley Nathaniel, Liya Ma
1. Connected Minds Trainee Retreat (York University), Toronto, ON 2025
- Repeated within-session intra- and extra-dimensional set shifts in the marmoset**
Authors: Marium Alvi, Ryley Nathaniel, Karmen Rai, Liya Ma
1. Society for Neuroscience (SfN) Conference, Chicago, USA 2024
- Rule-based flexible decision-making in rats**
Authors: Ryley Nathaniel, Yiping Zhang, Liya Ma
1. Natural Sciences and Engineering Research Council of Canada (NSERC), Toronto, ON 2023

Invited Talks

- Insights and challenges of using eye-tracking in marmoset research** **November 2025**
Presented at VPixx Technologies (vpixx.com)
Invited to speak about my research - which uses VPixx Technologies - and provide constructive feedback to the company's scientists and engineers.
- Becoming a graduate student** **October 2025**
Presented at York University
Invited to speak about my experience applying for and completing graduate school. Presenting

to fourth year undergraduate students interested in pursuing graduate studies.